

Key

1. (6 pts) Use the dictionary below to translate between English and formal sentential logic statements.

F: X is a field. G: X is a group. A: X is associative. C: X is commutative.

- a) X is a group but is not a field.

$$G \wedge (\neg F) \quad \text{or} \quad (G \wedge (\neg F))$$

- b) X is both associative and commutative when X is a field.

$$F \rightarrow (A \wedge C) \quad \text{or} \quad (F \rightarrow (A \wedge C))$$

2. (6 pts) Use a truth table to decide if the statement $[(p \vee q) \wedge (\neg p)] \rightarrow q$ is a tautology, a contradiction, or a contingency.

p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge (\neg p)$	$(p \vee q) \wedge (\neg p) \rightarrow q$
T	T	T	F	F	T
T	F	T	F	F	T
F	T	T	T	T	T
F	F	F	T	F	T

This is always
True so is
a tautology

3. (2 pts) What is the contrapositive of the statement $(\neg p) \rightarrow q$?

$$(\neg q) \rightarrow \neg(\neg p) \quad \text{or} \quad (\neg q) \rightarrow p \quad \text{or just} \quad \neg q \rightarrow p$$

4. (3 pts) Rewrite the expression $p \wedge (\neg q)$ as an equivalent expression using only the connectives $\sim$ and $\vee$

$$p \wedge (\neg q) \equiv \sim (\sim p) \vee (\sim(\neg q)) \equiv \sim (\sim p) \vee q$$

5. (3 pts) Rewrite the expression $p \leftrightarrow q$ as an equivalent expression using disjunctive normal form.

$$p \leftrightarrow q \equiv (p \wedge q) \vee ((\neg p) \wedge (\neg q))$$

T
F
F
T